

IP73-14: Urologist-Level Variation in Androgen Deprivation Therapy Intensification in Patients with Metastatic Hormone-Sensitive Prostate Cancer (mHSPC): A Retrospective, Population-Based Cohort Study in Ontario, Canada

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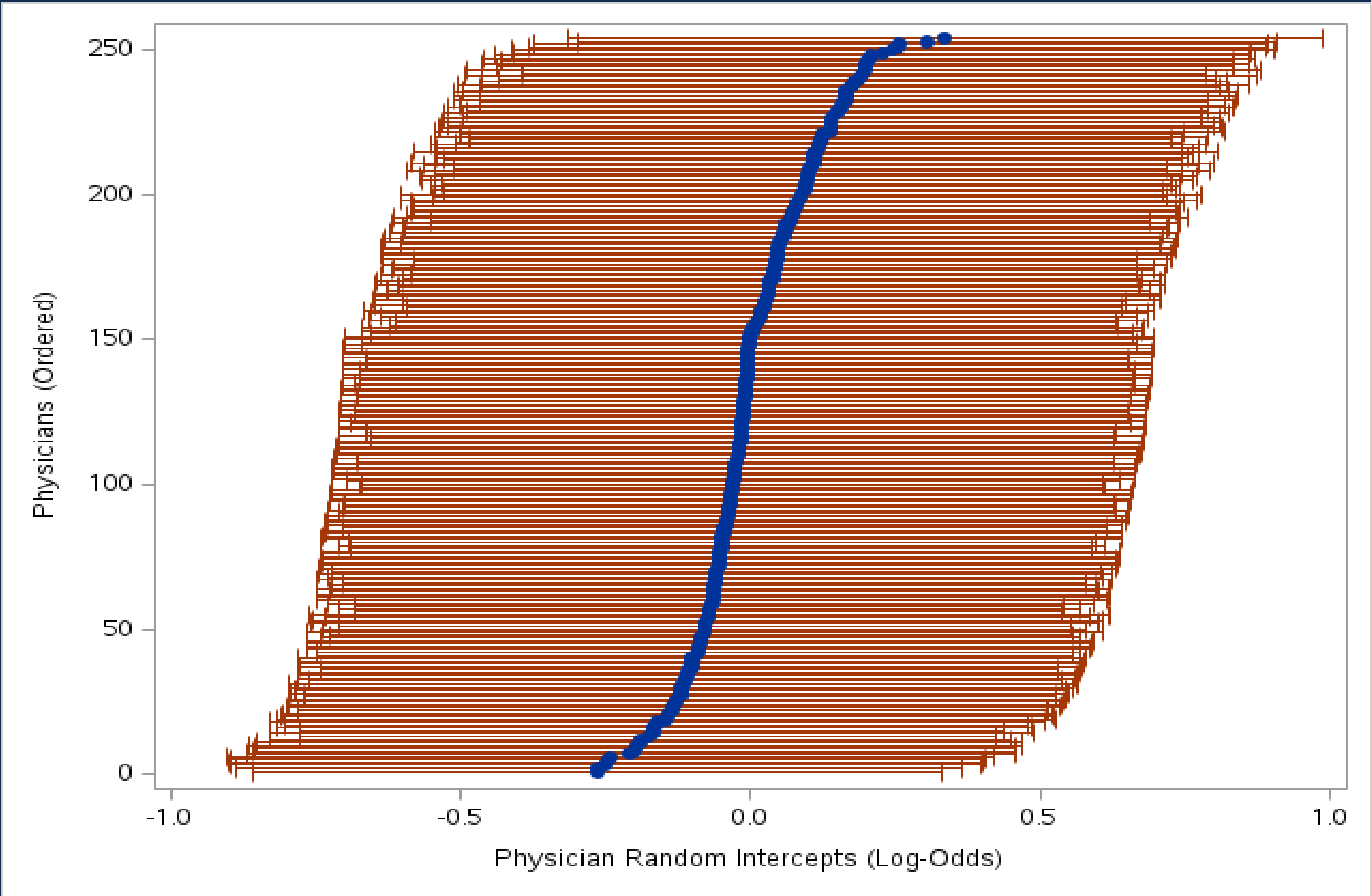
Background

- Guidelines recommend treatment intensification with ARPIs/docetaxel for men with de novo mHSPC, yet **real-world uptake remains low**
- Urologists frequently initiate ADT and strongly influence early treatment decisions, but it is unclear whether **variation in intensification reflects physician practice or patient characteristics**.
- In a universal health-care system, quantifying physician- versus patient-driven variation is essential to identify targets for improving equitable, guideline-concordant care

Methods

- Retrospective population-based cohort of men ≥ 66 years with de novo mHSPC in Ontario (2015–2022) whose ADT was initiated by a urologist
- Outcome:** receipt of treatment intensification (ARPI or docetaxel) within 6 months, attributed to the initiating urologist
- Analysis:** multilevel logistic regression clustered by physician with variance partition coefficient (VPC) to quantify physician-level contribution

A: 2015-2018



B: 2019-2022

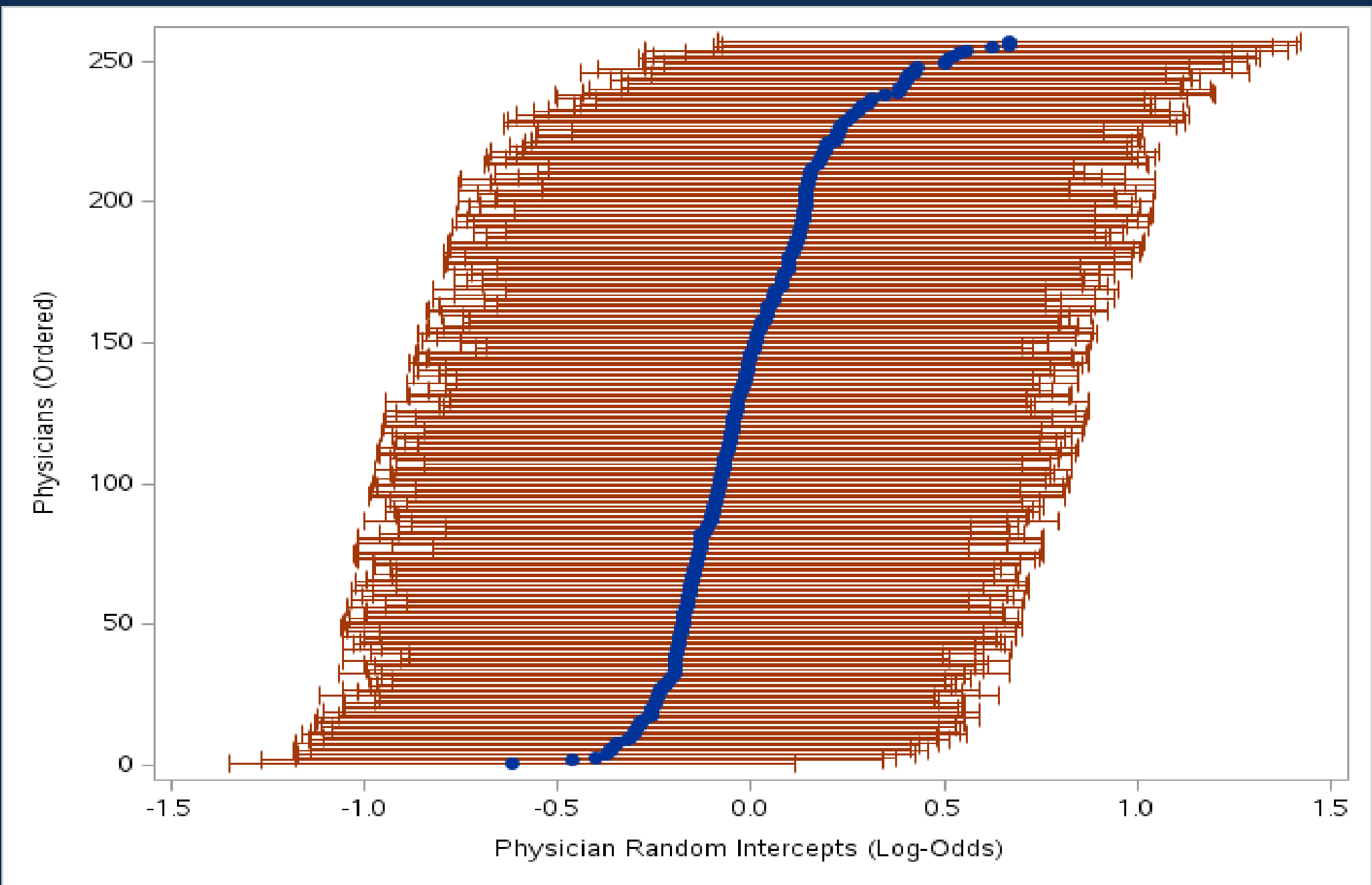


Figure 1. Physician-level variation in ADT intensification. Each point represents a urologist's odds for treatment intensification. Physician-level clustering explained **4.6%** of variation overall (2015–2022) and **7.8–8.5%** in 2020–2022

Results

- Among 3,871 patients treated by 332 urologists, **only 26.5% received treatment intensification** (median per urologist 21%, IQR 0–33%)
- Intensification increased over time, with mean uptake per urologist rising from **6% in 2014 to 59% in 2022**
- Physician-level factors explained little variation overall (VPC 4.6%), increasing modestly in 2020–2022 (VPC 7.8–8.5%), with **most variation driven by patient characteristics**

Future Directions

- Identify and address **patient-level barriers** (age, comorbidity burden, frailty) limiting uptake of treatment intensification
- Develop and test **system-level interventions**, including standardized referral pathways, clinical decision-support tools, and quality metrics
- Evaluate whether physician-level variation continues to evolve with **newer therapies, drug funding** and in **younger or non-urologist-managed populations**